

Chapter 2: Consumer Theory

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Review of EC1002

To introduce the foundational idea behind demand, consider an economy where there are many goods. A consumer needs to decide how much to purchase of each good.

Formally, there is a consumption bundle $\mathbf{x}$ where:

$$\mathbf{x} = (x_1, x_2, \dots, x_n)$$

where x_i denotes the quantity that a buyer wants to consume for good i . For example, x_1 represents 5 apples, x_2 is 5 litres of fuel, x_3 is 2 pieces of clothing, etc.

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In economics, we usually analyse how a consumer makes a decision between two goods. Therefore, consider **another** consumption bundle **y** where:

$$\mathbf{y} = (y_1, y_2, \dots, y_n)$$

where similarly, y_1 is 10 apples, y_2 is 3 litre of fuel, y_3 is 1 piece of clothing, etc.

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An important part of this chapter is to understand *why* consumers might prefer to choose one consumption bundle over the other.

The implication is that if we can model this, then we can get an insight as to the behavioural decision making of consumers in an economy.

To begin, we make some very basic assumptions about preferences.

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Completeness

The consumer has complete information about each bundle. This allows all possible bundles **x**, **y**, **z**, etc. to be comparable to each other.

For example, consider the three bundles below for various quantities of good a and b:

Table: Complete Bundles

Bundle	a	b
x	1	3
y	2	1
z	3	2

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Completeness simply means that a consumer sees all the values in that table. If this information was incomplete:

Table: Incomplete Bundles

Bundle	a	b
x	1	3
y	-	1
z	3	-

This prevents the consumer from being able to compare **x** with the other bundles, violating completeness.

Transitivity

This means that consumers can *rank* bundles an internally consistent way. For example, if $\mathbf{x}$ is preferred over $\mathbf{y}$ and $\mathbf{y}$ is preferred $\mathbf{z}$, then it must mean that $\mathbf{x}$ is preferred over $\mathbf{z}$.

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Monotonicity

This is also called "more is preferred to less". The basic implication is that if there are two bundles **x** and **y** that are identical **except** that **x** contains one unit extra of a good, then it is preferred over **y**.

For example, consider the two bundles for 5 goods:

Table: Monotonicity

Bundle	Good				
	a	b	c	d	e
x	1	2	6	3	4
y	1	2	6	3	3

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This is a very technical assumption which basically implies that consumers prefer bundles that contain average quantities of goods compared to extreme quantities.

For example, consider the two bundles for 4 goods:

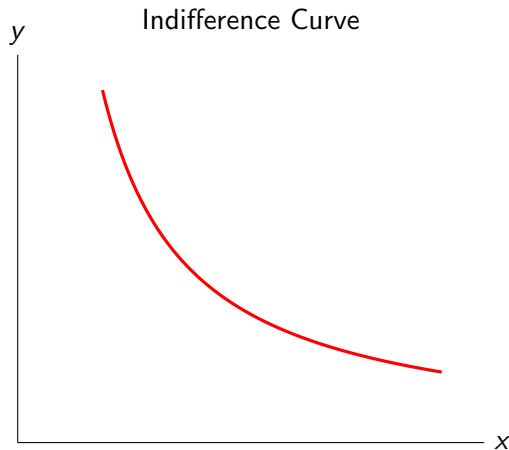
Table: Convexity

Bundle	Good			
	a	b	c	d
x	5	5	5	5
y	0	0	0	20

In this case, convexity would mean that a consumer prefers **x** over **y**.

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With these assumptions established, we can now construct **indifference curves**. This takes the below form:



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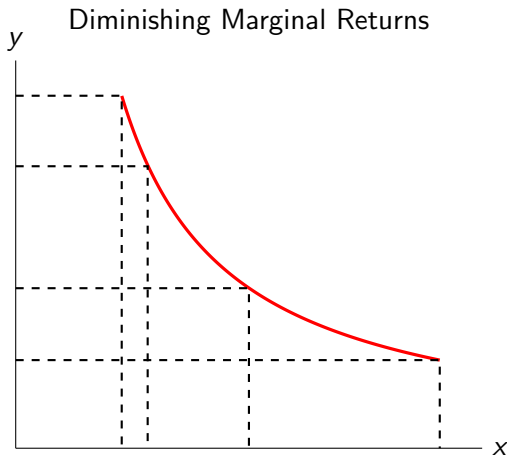
You should note that the slope of an indifference curve is given by:

$$\frac{\partial u}{\partial x} / \frac{\partial u}{\partial y} = MRS,$$

which is called the Marginal Rate of Substitution.

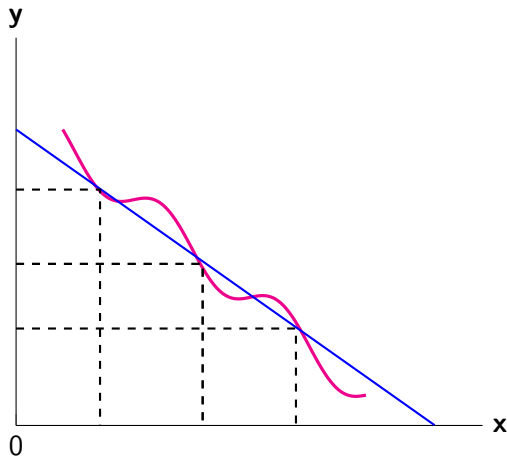
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An interesting property about the MRS is that it is **diminishing**. This comes from the **convexity assumption**:



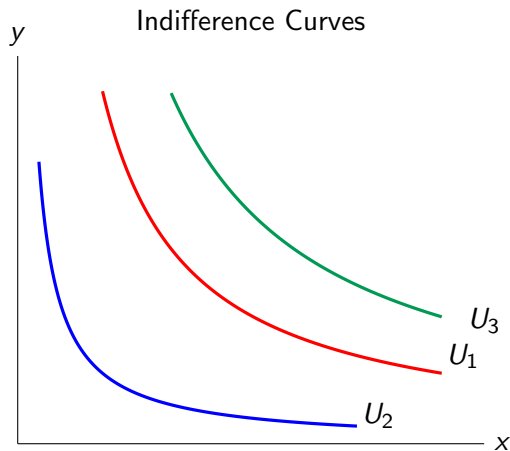
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For optimisation, (strict) convexity ensures the existence of a global maximum. If there is no convexity, then we can have multiple points of tangency:



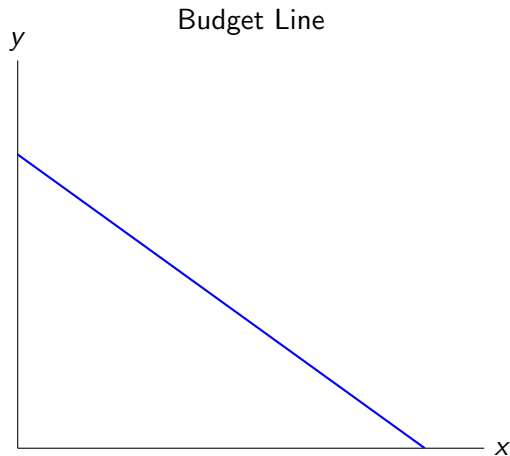
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The monotonicity assumption ensures that a consumer always prefers to have a higher utility level. Geometrically, this means the consumer is more satisfied the further away they are from the origin:



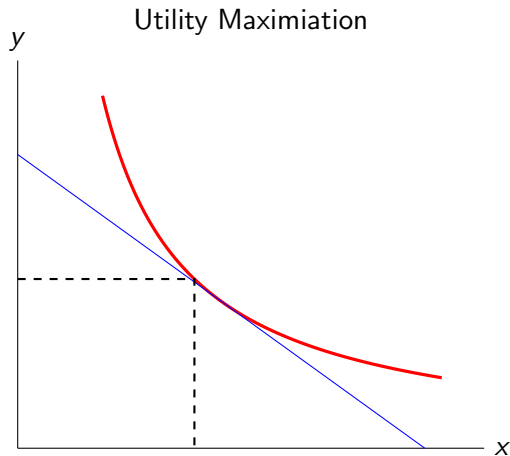
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A consumer also has a budget constraint $M = p_x X + p_y Y$, which is a linear function that can be plotted:



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We will discuss in this chapter how a consumer maximises their utility subject to their budget constraint, which leads to the following optimal point:



1 Indifference Curves and the Budget Constraint

This chapter goes over the concepts discussed in Block 4 and 9 of EC1002.

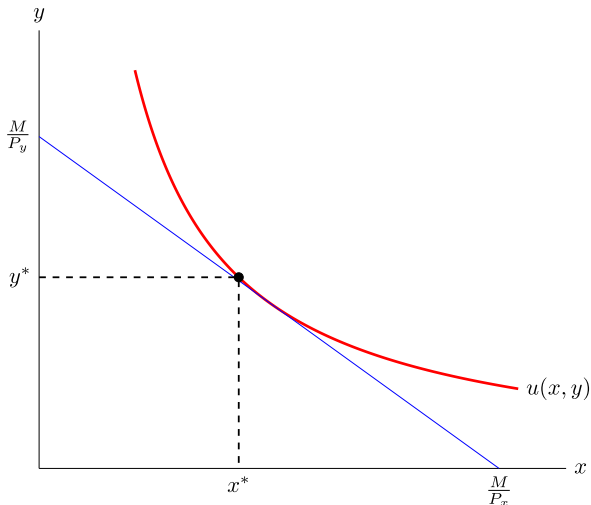
The main theme is that we develop a theory of how individuals behave in a hypothetical economy.

The principle objective function that an individual solves is to maximise their satisfaction (utility function) subject to their budget constraints:

$$\max u(x, y) \quad \text{s.t.} \quad P_x x + P_y y = M$$

1 Indifference Curves and the Budget Constraint

We learned that this leads to the following graphical solution:



1 Indifference Curves and the Budget Constraint

That is, the optimal point that maximises utility is when the slope of the indifference curve (also called the marginal rate of substitution) is equal to the slope of the budget constraint:

$$MRS = \frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

In this module, we need to derive this result mathematically. The starting point is the objective function and its constraint:

$$\max U(x, y) \quad \text{s.t} \quad p_x x + p_y y = M$$

To solve this, we need to use the Lagrangian:

$$\mathcal{L} = u(x, y) + \lambda(M - P_x x - P_y y)$$

1 Indifference Curves and the Budget Constraint

Taking the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial u}{\partial x} - \lambda P_x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial u}{\partial y} - \lambda P_y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - P_x x - P_y y = 0.$$

From the first two conditions:

$$\frac{\partial u}{\partial x} \frac{1}{P_x} = \frac{\partial u}{\partial y} \frac{1}{P_y}$$

$$\frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = \frac{P_x}{P_y}$$

$$\frac{MU_x}{MU_y} = MRS = \frac{P_x}{P_y},$$

which satisfies the tangency condition that leads to the graphical solution

For an explicit utility function, it's possible to derive the optimal values of x and y when subject to the budget constraint.

1 Indifference Curves and the Budget Constraint

Example 2.1

Example 2.1: A consumer has the following Cobb–Douglas utility function:

$$u(x, y) = x^{1/2}y^{1/2}$$

where $\alpha, \beta > 0$. The price of x is P_x and the price of y is P_y . The consumer's income is M . Derive the demand functions for x and y .

1 Indifference Curves and the Budget Constraint

Example 2.1

Solution: The consumer's problem is:

$$\max_{x,y} x^{1/2}y^{1/2} \quad \text{s.t.} \quad P_x x + P_y y = M$$

Using the Lagrange multipliers method:

$$\mathcal{L} = x^{1/2}y^{1/2} + \lambda(M - P_x x - P_y y)$$

1 Indifference Curves and the Budget Constraint

Example 2.1

Deriving the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{x^{-1/2}y^{1/2}}{2} - \lambda P_x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{x^{1/2}y^{-1/2}}{2} - \lambda P_y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - P_x x - P_y y = 0.$$

1 Indifference Curves and the Budget Constraint

Example 2.1

From the first two equations, we start by making y the subject:

$$\frac{x^{-1/2}y^{1/2}}{2P_x} = \frac{x^{1/2}y^{-1/2}}{2P_y}$$

$$\frac{y^{1/2}}{y^{-1/2}} = \frac{2P_x}{2P_y} \frac{x^{1/2}}{x^{-1/2}}$$

$$y = \frac{P_x}{P_y}x$$

1 Indifference Curves and the Budget Constraint

Example 2.1

To find the optimised value of x , we substitute y into the budget constraint and make x the subject:

$$M = P_x x + P_y y$$

$$M = P_x x + P_y \frac{P_x}{P_y} x$$

$$M = P_x x + P_x x$$

$$x^* = \frac{M}{2P_x}$$

1 Indifference Curves and the Budget Constraint

Example 2.1

The optimised value of y is therefore:

$$y = \frac{P_x}{P_y} x$$

$$y = \frac{P_x}{P_y} \frac{M}{2P_x}$$

$$y^* = \frac{M}{2P_y}$$

Which means that the value of utility at the optimum is:

$$u^* = x^{1/2} y^{1/2}$$

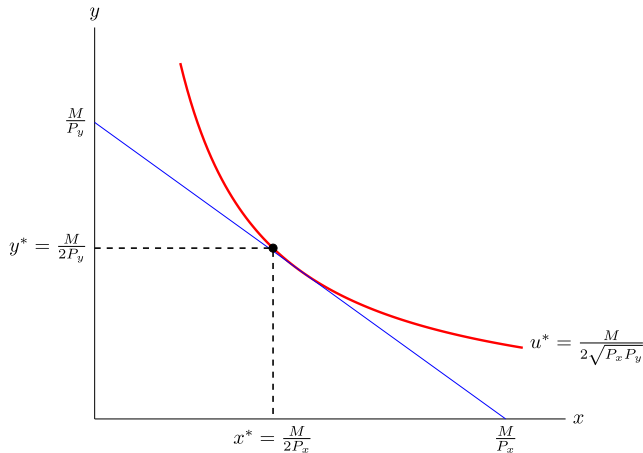
$$= \left(\frac{M}{2P_x} \right)^{1/2} \left(\frac{M}{2P_y} \right)^{1/2}$$

$$= \frac{M}{2\sqrt{P_x P_y}}$$

1 Indifference Curves and the Budget Constraint

Example 2.1

Graphically, this can be shown as:



2 Adjustment to Prices

We discussed that when a price of a good changes, then a consumer theoretically chooses a new consumption bundle that incorporates the substitution and income effect.

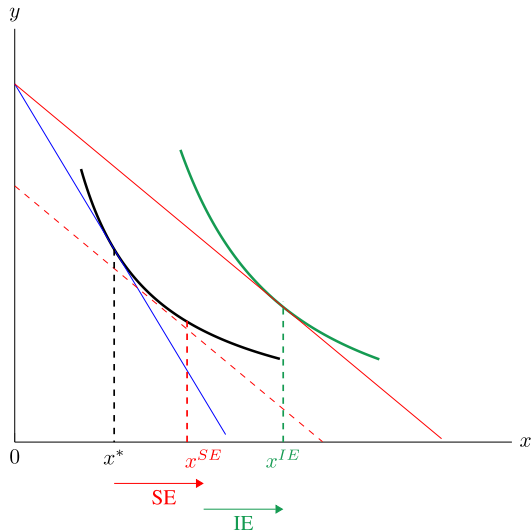
In EC1002, this was calculated using the Hicksian decomposition.

In this module, we demonstrate this method mathematically.

2.1 The Hicksian Decomposition

Following a change in price, the Hicksian decomposition asks "how many goods would the consumer purchase at the new price but at their old utility curve".

For a normal good, we can show this graphically as:



2.1 The Hicksian Decomposition

Example 2.2

Let us now derive this result mathematically:

Example 2.2: Suppose $u(x, y) = x^{1/2}y^{1/2}$. Income is at $M = 72$, the price of y is set at $P_y = 1$ and the price of x is initially set at $P_x = 9$ but reduces to $P'_x = 4$. Calculate the total price effect of x using the Hicksian decomposition.

2.1 The Hicksian Decomposition

Example 2.2

Solution:

For the utility function $u(x, y) = x^{1/2}y^{1/2}$ and budget constraint $P_x x + P_y y = M$, we have derived the solution in Example 2.1. To keep things simple, we only focus on x :

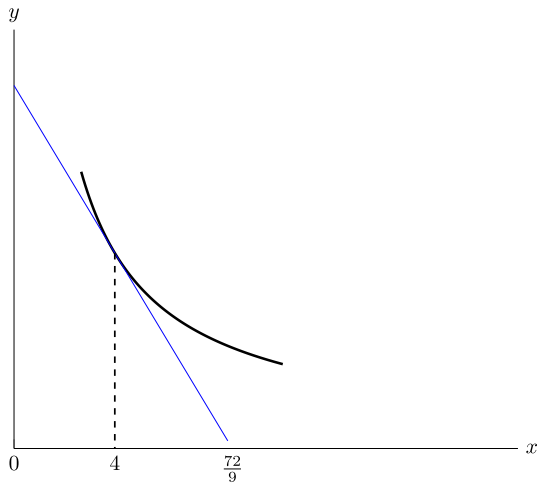
$$x^* = \frac{M}{2P_x} = \frac{72}{2 \cdot 9} = 4$$

where the optimal utility is a constant given as:

$$u^* = \frac{M}{2\sqrt{P_x P_y}} = \frac{72}{2\sqrt{9 \cdot 1}} = 12$$

2.1 The Hicksian Decomposition

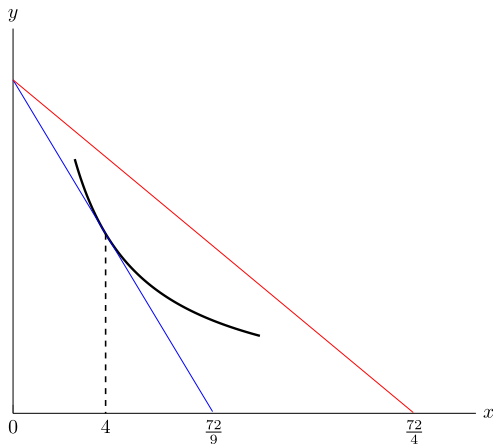
Graphically, this would be shown as:



2.1 The Hicksian Decomposition

Example 2.2

Following the price decrease to $P'_x = 4$, we can calculate the new budget constraint as $\frac{M}{P'_x} = \frac{72}{4} = 18$:

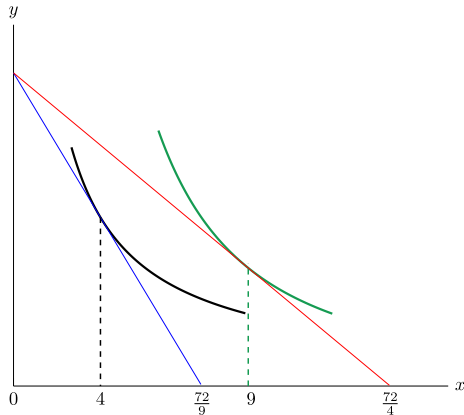


2.1 The Hicksian Decomposition

Example 2.2

At the new price level, the consumer will optimally select:

$$x'^* = \frac{M}{2 \cdot P'_x} = \frac{72}{2 \cdot 4} = 9$$



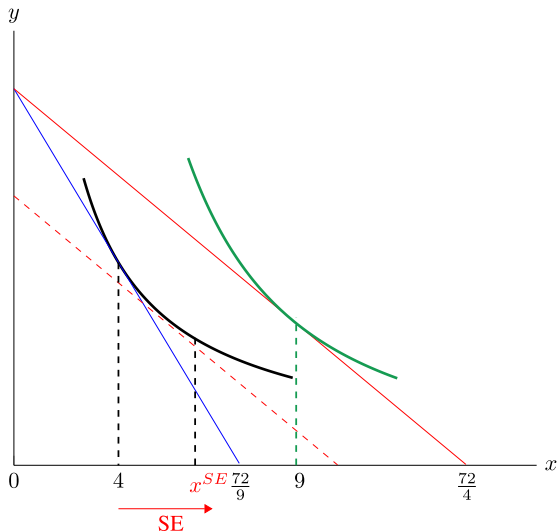
This means that the total price effect (PE) is $9 - 4 = 5$. However, the 9 includes both income and substitution effects. We need a way to decompose it from this combined amount.

2.1 The Hicksian Decomposition

Example 2.2

We calculate the substitution effect using the Hicksian method.

Graphically, we draw a budget line that has the same slope as the new budget line, but make it tangent to the old utility curve:



2.1 The Hicksian Decomposition

Example 2.2

Mathematically, we know that the optimal point is when $x^* = \frac{M}{2P_x}$. This means that x^{SE} can be described as:

$$x^{SE} = \frac{M - \Delta M}{2P'_x},$$

where the ΔM component is called the **compensating variation**. It is defined as the amount of money that we need to take away^a in order to bring the consumer back to the original utility curve. In order to derive the explicit value of ΔM , you need to note that the optimal utility at the substitution effect is:

$$u^{SE*} = \frac{M - \Delta M}{2\sqrt{P_x P_y}}$$

^aIt is also the amount of money that we *give* in the instance of a price increase.

2.1 The Hicksian Decomposition

Example 2.2

However, according to the Hicksian decomposition, we know that $u^{SE*} = u^*$. That is, they are both on the same utility curve (you can see this in the graph). Therefore, we can use this result to derive the compensating variation ΔM :

$$\frac{M - \Delta M}{2\sqrt{P'_x P_y}} = u^*$$

$$\frac{M - \Delta M}{2\sqrt{P'_x P_y}} = \frac{M}{2\sqrt{P_x P_y}}$$

$$\frac{72 - \Delta M}{2\sqrt{4 \cdot 1}} = \frac{72}{2\sqrt{9 \cdot 1}}$$

Therefore: $\Delta M = 24$

2.1 The Hicksian Decomposition

Example 2.2

Substituting this into x^{SE} :

$$\begin{aligned}x^{SE} &= \frac{M - \Delta M}{2P'_x} \\&= \frac{72 - 24}{2 \cdot 4} = 6\end{aligned}$$

We can therefore decompose the price effect from the income and substitution effect:

$$PE = (SE + IE) = 9$$

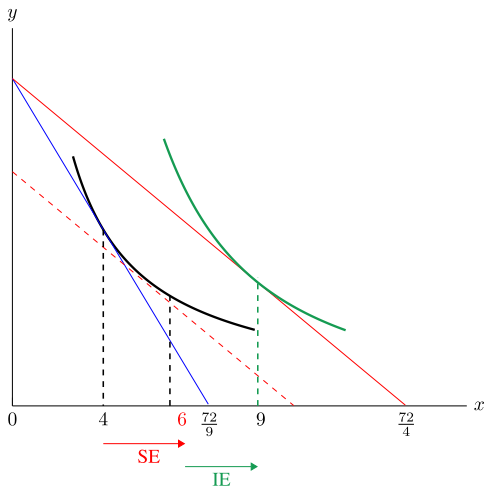
$$= (6 + IE) = 9$$

Which means $IE = 3$

2.1 The Hicksian Decomposition

Example 2.2

Graphically:



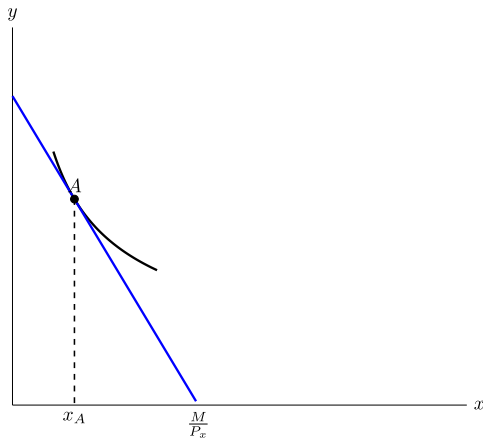
2.2 The Equivalent Variation

In the previous section, we introduced the concept of compensating variation. Following a price adjustment, this is the **amount of income that we would change** from the consumer in order to restore them back to the **original utility function**, but at the **new price ratio**.

A related concept in microeconomic theory is called the equivalent variation, which is essentially the reverse. Following a price adjustment, this is the **amount of money that we would change** from the consumer in order to keep them at the **new utility curve**, but at the **old price ratio**.

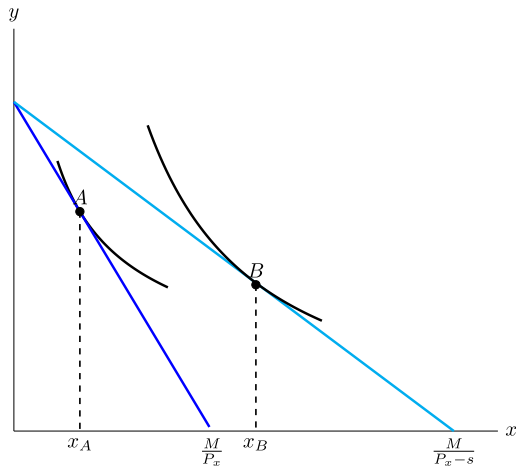
2.2 The Equivalent Variation

There are important policy considerations based on how we model behaviour in an economy. Consider an individual who has a budget M set for spending between food and fuel. For a given utility function, we can illustrate their optimal choice as:



2.2 The Equivalent Variation

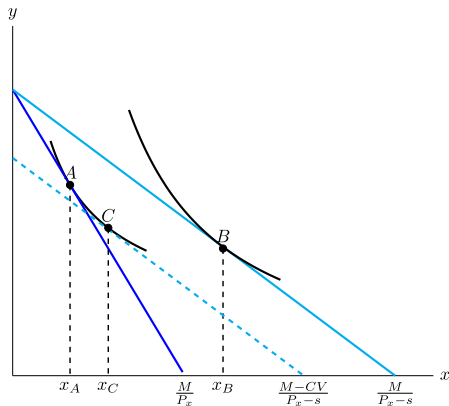
Now suppose that the government wants to intervene and offer a subsidy s for fuel to consumers. This policy would have the same effect of a price decrease of fuel $P_x - s$:



2.2 The Equivalent Variation

The total price effect would be the movement from A to B .

If we were to illustrate the substitution effect using the Hicksian decomposition, we would calculate the compensating variation to restore the consumer to the original utility curve but at the new price ratio as shown on the right:



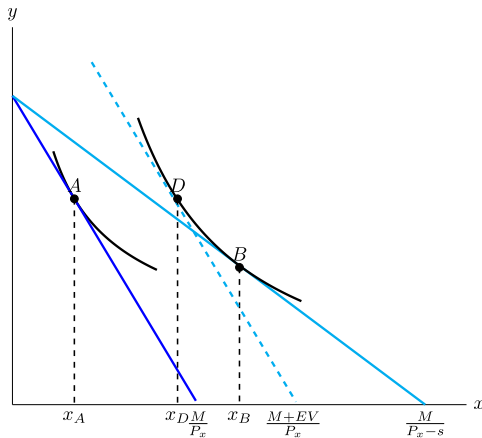
2.2 The Equivalent Variation

Overall, we can clearly see that the subsidy policy is good for the consumer. This is because they will choose the point B which is on a higher utility curve than on A .

It is interesting to note that the government is currently spending $s \cdot x_B$ amount of money in order to implement the policy. This raises an important question as to whether we can achieve the *same outcome* of consumer satisfaction but at a lower cost to the government.

2.2 The Equivalent Variation

Consider that instead of giving consumers a subsidy s (so the price of heating is back to P_x), the government transfers a lump-sum amount of money that keeps the consumer on *the new utility curve*:



2.2 The Equivalent Variation

Notice that even though at D the individual is consuming less x compared to point D they are still on the same level of utility as the subsidy policy. The lump-sum amount of money that achieves this specific outcome is known as the equivalent variation.

The policy of transferring an equivalent variation is cheaper on the government compared to the subsidy option. One way to see this is that if the second option was implemented, then point B is no longer feasible as it is geometrically on a higher budget constraint.

The intuitive difference between the two policies is that the subsidy creates a substitution effect on the consumer's decision making.

2.2 The Equivalent Variation

The fact that only fuel is cheaper would mean the consumer ends up purchasing more of it^b. On the other hand, an equivalent variation in income would lead the consumer to more equally distribute their spending behaviour between both x and y .

An important point to keep in mind is that when the price of a good **decreases**, it means that the compensating variation **deducts** from the income, whereas the equivalent variation **adds** to the income.

You must keep in mind that if the price of a good **increases**, then the signs will reverse. The next example highlights this.

^bConsider a subsidy on electricity. This usually leads to individuals becoming less careful about the amount of electricity they use in their homes. It ultimately leads to a greater increase in usage and increases the costs on the government.

2.2 The Equivalent Variation

Example 2.6

Example 2.6: Suppose a consumer has the utility function $u(x, y) = x^{1/2}y^{1/2}$. For a budget constraint $M = P_x x + P_y y$, we know that this leads to the consumption points:

$$x^* = \frac{M}{2P_x} \quad y^* = \frac{M}{2P_y}$$

Which leads to the optimal utility level:

$$u^* = \frac{M}{2\sqrt{P_x P_y}}$$

Let us say that $M = 100$, with the price of y $P_y = 1$ and the price of x increases from $P_x = 1$ to $P'_x = 2$.

Focusing just on good x , calculate and show graphically the Hicksian decomposition (which clearly shows the compensating variation amount). After that, calculate and illustrate a policy of equivalent variation following the price increase.

2.2 The Equivalent Variation

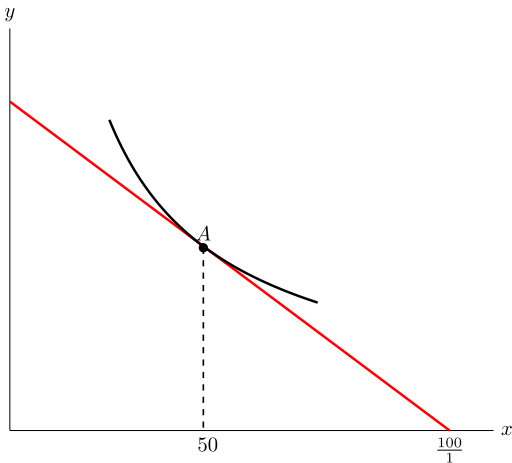
Example 2.6

Solution:

Let us calculate the initial point of x^A

This is simply:

$$x^A = \frac{M}{2P_x} = \frac{100}{2} = 50$$



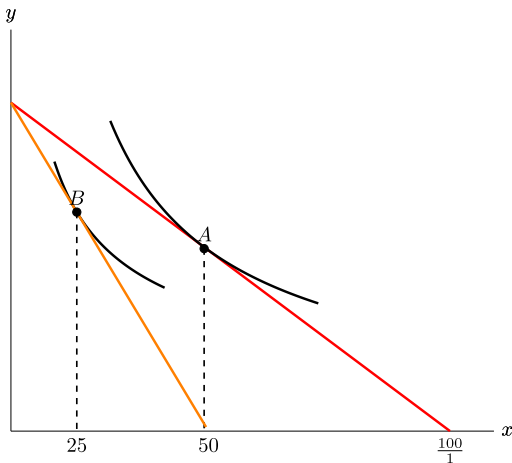
2.2 The Equivalent Variation

Example 2.6

Following the price increase, the budget line rotates inwards.

The new optimal point is:

$$x^B = \frac{M}{2P'_x} = \frac{100}{2 \cdot 2} = 25$$



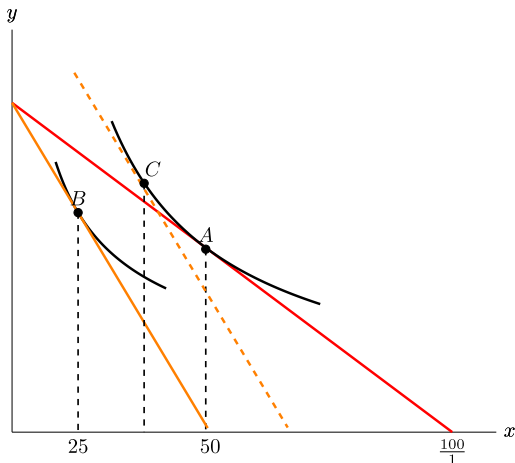
2.2 The Equivalent Variation

Example 2.6

Therefore, the movement from point A to B represents the total price effect.

We now need to calculate the substitution effect using the Hicksian decomposition. That is, we want to calculate the level of consumption at the original utility curve but with the new price ratio:

$$x^C = \frac{M + CV}{2P'_x}$$



2.2 The Equivalent Variation

Example 2.6

We do not have the compensating variation value $CV = \Delta M$ just yet (notice that this is an addition given that we are illustrating a price *increase*). Since the original bundle A and substitution amount C are in the same utility curve, then we can say that:

$$u^C = u^A$$

$$\frac{M + CV}{2\sqrt{P'_x P_y}} = \frac{M}{2\sqrt{P_x P_y}}$$

$$\frac{100 + CV}{2 \cdot \sqrt{2 \cdot 1}} = \frac{100}{2 \cdot \sqrt{1 \cdot 1}}$$

$$CV = 41.42$$

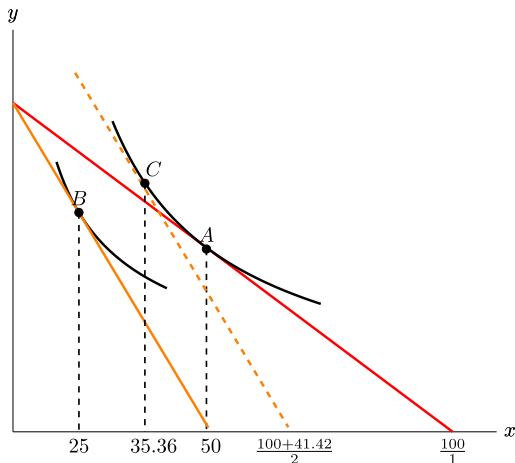
2.2 The Equivalent Variation

Example 2.6

Where $CV = 41.42$ is the compensating variation.

This means that x^C can be calculated as

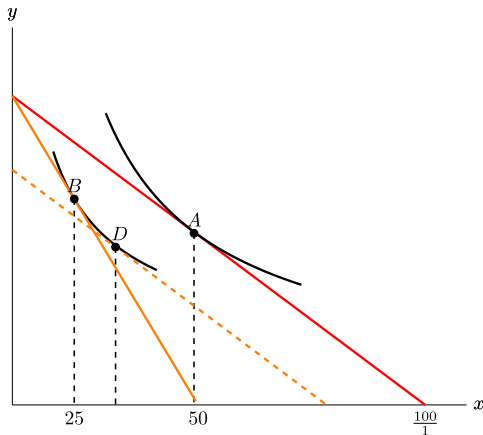
$$\begin{aligned}x^C &= \frac{M + CV}{2P'_x} \\&= \frac{100 + 41.42}{2 \cdot 2} = 35.36\end{aligned}$$



2.2 The Equivalent Variation

Example 2.6

Now let us consider a policy of equivalent variation. That is, we want to keep the consumer at the new utility curve but using the old price ratio:



2.2 The Equivalent Variation

Example 2.6

We can describe point the consumption of point D as:

$$x^D = \frac{M - EV}{2P_x}$$

Just as before, we need to derive the equivalent variation value. To do this, notice that B and D are in the same utility curve. This means that:

$$u^B = u^D$$

$$\frac{M}{2\sqrt{P'_x P_y}} = \frac{M - EV}{2\sqrt{P_x P_y}}$$

therefore, $EV = 29.29$

$$\frac{100}{2\sqrt{2 \cdot 1}} = \frac{100 - EV}{2\sqrt{1 \cdot 1}}$$

3 The Intertemporal Consumption Choice

Most of the economic analysis conducted at the introductory level is static.

This means that we only observe equilibrium changes in a single time period.

In order to examine higher level models, we can extend our analysis dynamically across time.

3.1 Model Setup

Consider an economy which exists only for two time periods $t = 1, 2$. In the first period, an individual earns Y_1 but also spends C_1 . We can define the difference as:

$$Y_1 - C_1 = S$$

We can see that if $Y_1 > C_1$, then the individual will have money saved that they can use in the next period. Alternatively, if $Y_1 < C_1$, then they will need to borrow money from the bank.

If the individual has a positive S , then the bank will pay them $(1 + r)$ on the savings in the next period. If they have a negative S , then the individual needs to make sure that it is paid back with interest $(1 + r)$ the next period.

3.1 Model Setup

Now in the second period, the consumer also earns Y_2 and must also decide on how much to spend C_2 .

The intertemporal budget constraint can therefore be defined as:

$$C_2 = Y_2 + (1 + r)S$$

$$C_2 = Y_2 + (1 + r)(Y_1 - C_1)$$

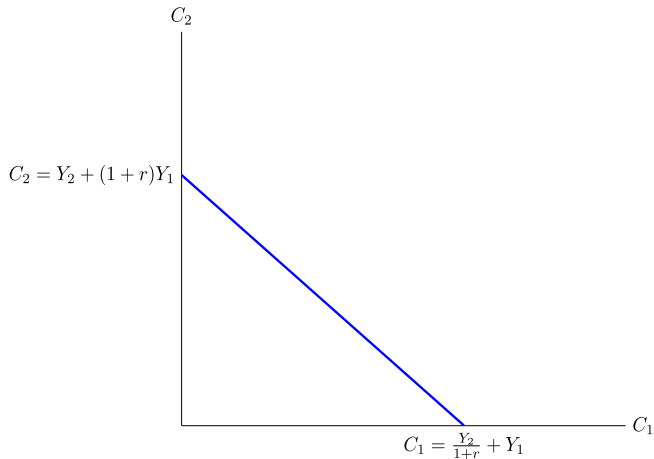
$$C_2 = Y_2 + (1 + r)Y_1 - (1 + r)C_1$$

3.1 Model Setup

Let us plot C_2 on the vertical axis and C_1 on the horizontal axis.

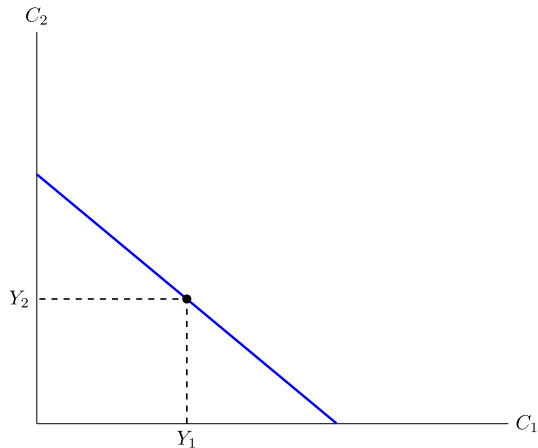
This is a downward sloping linear equation:

$$\frac{\partial C_2}{\partial C_1} = -(1+r)$$



3.1 Model Setup

We can illustrate the income endowments Y_1 and Y_2 at:



3.1 Model Setup

Now we need to translate how a consumer decides to become a saver or a borrower.

Essentially, this is a decision that depends on each individual. We can think of their choice of C_1 and C_2 in the same way as they decide between two goods x and y .

Formally, the consumer maximises their utility function of their choice (C_1, C_2) subject to their intertemporal budget constraint:

$$\max u(C_1, C_2) \quad \text{s.t.} \quad C_2 = Y_2 + (1 + r)Y_1 - (1 + r)C_1$$

Using the Lagrangian, we arrive at an identical solution:

$$\frac{MU_{C_1}}{MU_{C_2}} = 1 + r,$$

which means that the consumer selects a bundle when their utility curve is tangent to their budget constraint.

3.1 Model Setup

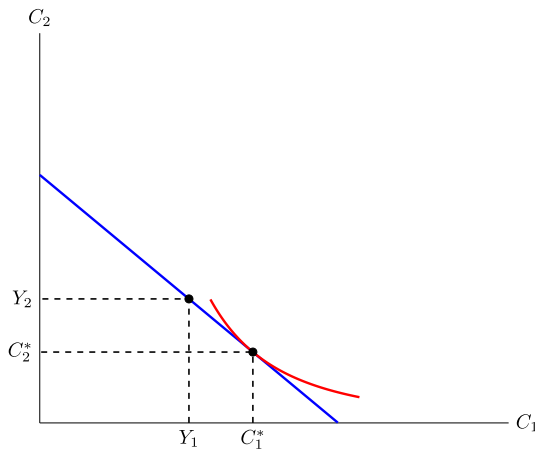
Using the Lagrangian, we arrive at an identical solution:

$$\frac{MU_{C_1}}{MU_{C_2}} = 1 + r,$$

which means that the consumer selects a bundle when their utility curve is tangent to their budget constraint.

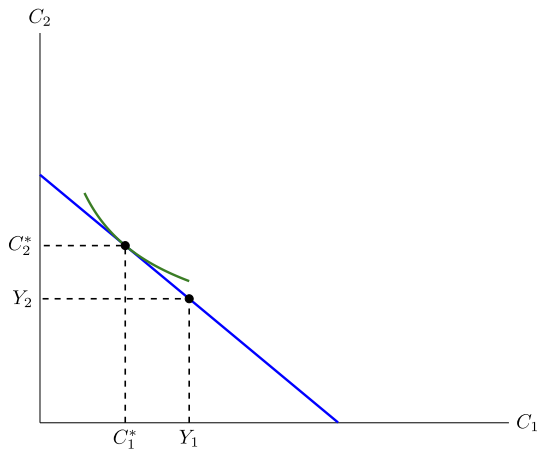
3.1 Model Setup

If the individual was a borrower, then they will choose a consumption combination where C_1 exceeds their first period income Y_1 :



3.1 Model Setup

Similarly, if the consumer was a saver, then they select a consumption combination where C_1 is less than their first period income Y_1 :



3.2 Adjustment to Interest Rate

Our analysis of the intertemporal consumption choice is identical to that of a consumer choosing between goods x and y .

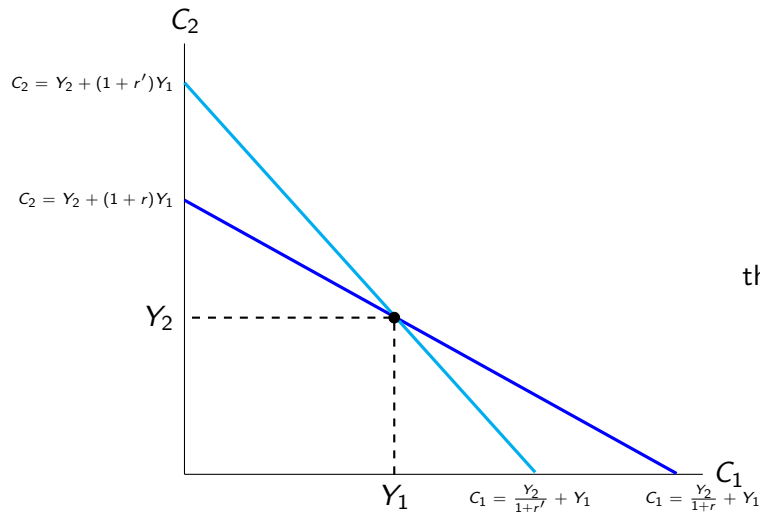
We saw that when the price of good x changes, the budget line rotates due to the horizontal intercept shifting.

In the intertemporal case, the price of consumption is the interest rate r and *both* intercept points share this variable.

Therefore, a change in interest rate *pivots* in both directions.

3.2 Adjustment to Interest Rate

For example, suppose the interest rate increases $r' > r$. This can be shown as:



Notice how the new budget line passes through the initial income position (Y_1, Y_2)

3.2 Adjustment to Interest Rate

Just like in the case of goods x and y , a change in the interest rate leads to a substitution and income effect on the new consumption choice.

However, we will see that the outcome differs based on whether the individual is a borrower or a saver.

3.2 Adjustment to Interest Rate

Example 2.A

Example 2.A: Consider an individual making an intertemporal consumption decision.

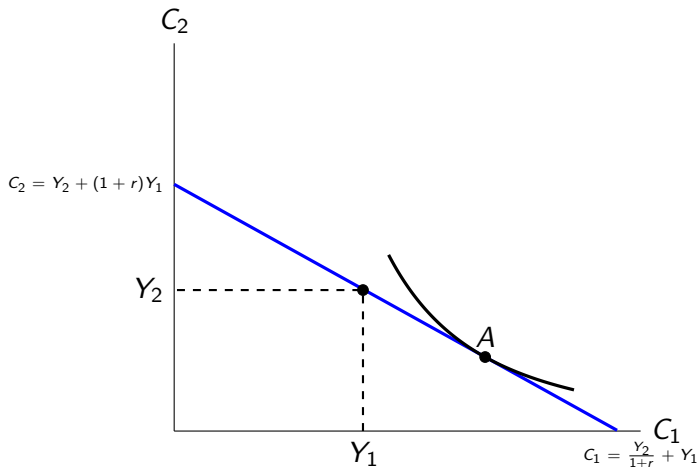
Illustrate their initial choice given that they are a borrower.

Demonstrate graphically the substitution and income effect that follows from an increase in interest rate.

3.2 Adjustment to Interest Rate

Example 2.A

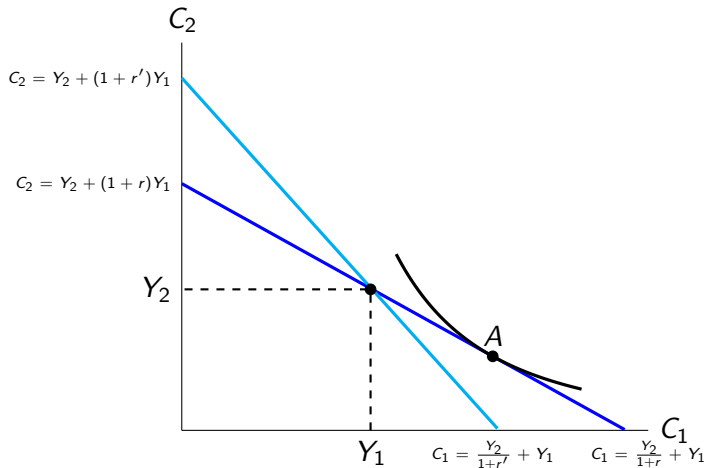
Solution: A borrower will select a point where their initial consumption C_1 exceeds their initial income Y_1 :



3.2 Adjustment to Interest Rate

Example 2.A

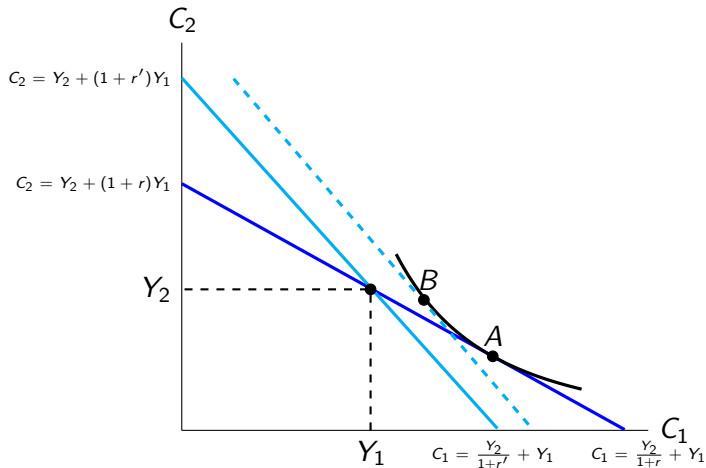
Following an increase in interest rate $r' > r$, the budget line pivots around the endowment point:



3.2 Adjustment to Interest Rate

Example 2.A

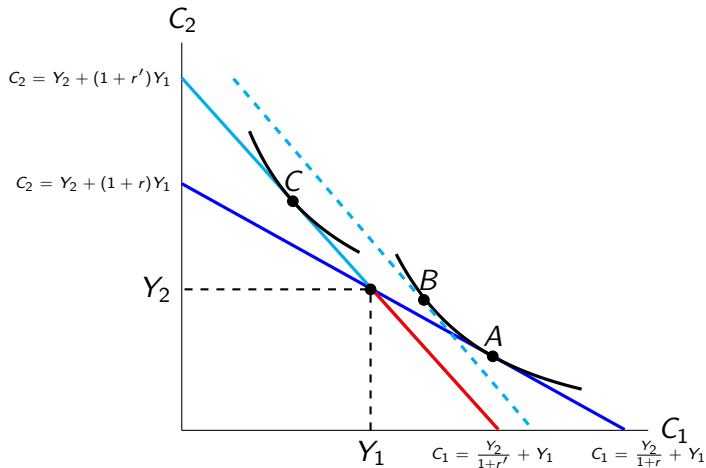
Just as we have learned before, the Hicksian substitution effect is demonstrated by restoring the individual to the original utility curve but with the new price ratio:



3.2 Adjustment to Interest Rate

Example 2.A

Geometrically, the individual will select a point which gives them a higher utility compared to their initial point. For example, they would choose point C:



3.2 Adjustment to Interest Rate

Example 2.A

We can see that if the individual selects any point on the red line, they are choosing to continue as a borrower.

However, this offers a lower utility compared to their initial position.

Intuitively, when interest rates increase, a borrower becomes poorer. This is because it is more expensive to take on debt.

Therefore, the increase in r will force the consumer to borrow less or to become a saver, such as at point C .

3.2 Adjustment to Interest Rate

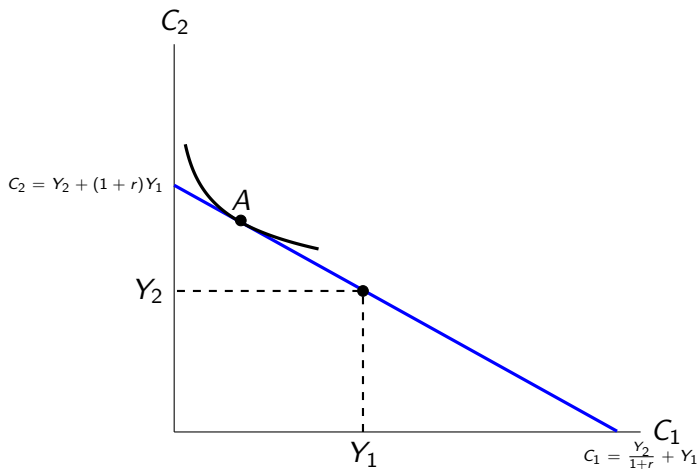
Example 2.B

Example 2.B: Now consider the same scenario of a rise in interest rate but the individual begins as a saver.

3.2 Adjustment to Interest Rate

Example 2.B

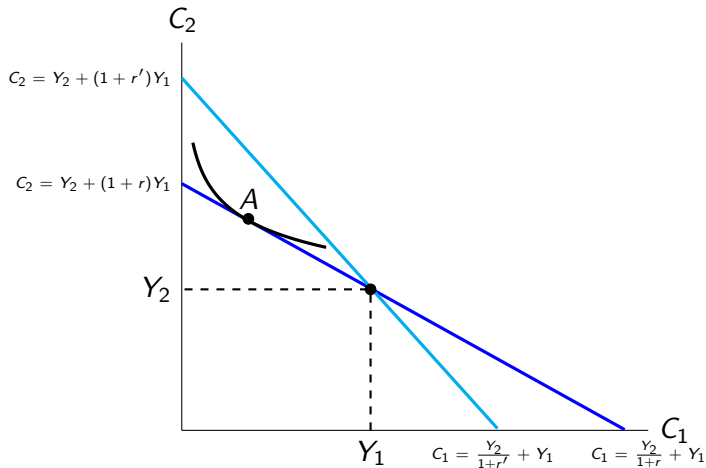
Solution: We begin with the individual choosing a point where $C_1 < Y_1$:



3.2 Adjustment to Interest Rate

Example 2.B

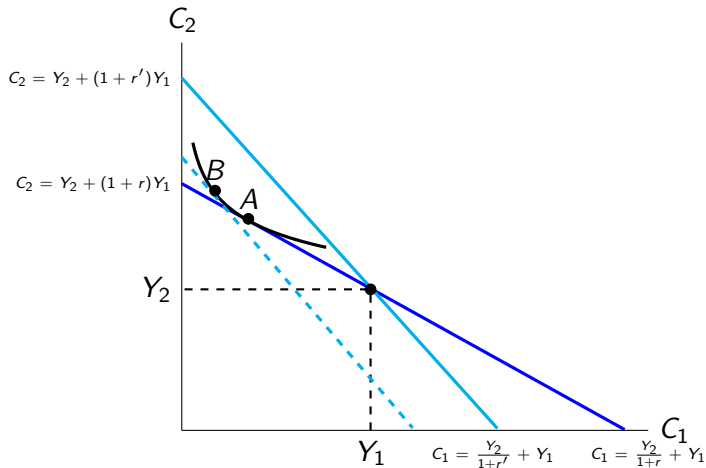
Following a rise in interest rates $r' > r$, the budget line rotates around the endowment point:



3.2 Adjustment to Interest Rate

Example 2.B

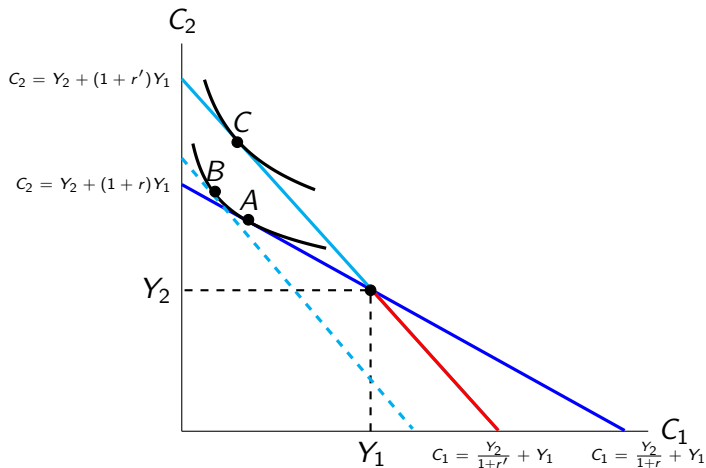
The Hicksian substitution effect would restore the individual to the original utility curve but with at the new price:



3.2 Adjustment to Interest Rate

Example 2.B

Just as the previous case, the individual can select any point to the left of the endowment position as it offers a higher utility to them. For example, they will select point C:



3.2 Adjustment to Interest Rate

Example 2.B

They will not select a point on the red line, as it offers a lower utility position compared to their initial consumption decision.

Intuitively, a rise in interest rate makes the saver richer (since they benefit from getting more money from their savings). However, the increase in income can be large enough that it would encourage them to spend more.

Essentially, it is theoretically ambiguous where C will lie (although it will certainly be to the left of the endowment point). Notice that unlike the previous case, the substitution and income effect go in opposite directions. If the net substitution effect becomes negative, then it implies that C_1 is an inferior good.